

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology, Kanpur	8.2/10
2020	CBSE(XII)	Mallikarjun School, Pithoragarh	97.2%
2018	CBSE(X)	Mallikarjun School, Pithoragarh	96.4%

Scholastic Achievements

- Secured **All India Rank 568** in **Joint Entrance Examination(Advanced) 2021** among 1.4 Lakh shortlisted candidates.
- Secured **All India Rank 6423** from the 9 lakh applicants in **Joint Entrance Examination(Mains) 2021**.

Competitive Programming

- Maximum rating 1912 (**Candidate Master**) on Codeforces. (Handle : yuvrajKharayat)
- Maximum rating 2115 (5 Stars) on CodeChef. (Handle: yuvraj3004)
- Maximum rating 1558 (Cyan) on Atcoder. (Handle: yuvrajKharayat)

Key Projects

- Volksmarkt** |  (Course Project) (Jan'23- Apr'23)
Course: Software Development and Organization
 - Developed a **full stack web app** for purchasing items from shops inside the campus.
 - Used **React.js** for frontend, with **Material UI** for responsive components and **Django** for backend.
 - Implemented requirements and design documents in a systematic manner using plan-driven development methodologies
- CFLockout** |  (Self Project) (Jun'23-Present)
 - Created a lockout-style coding **competition platform** inspired by Codeforces, leveraging the **Codeforces API**.
 - Used **Next.js** for the frontend, with **Material UI** for responsive components and **Zustand** for central state managment.
 - Engineered the backend server in **Golang**, utilizing the **gin gonic** web framework, and **PostgreSQL** as the database.
 - Implemented a secure token based **authentication** and **authorization** system using **JSON Web Tokens(JWT)**.
 - Utilized **go routines** to achieve **concurrency** for handling multiple sessions, optimizing the platform's performance.
- CSE Bubble** |  (Course Project) (Jan'23-Apr'23)
Course: Computer Organization
 - Developed a **32-bit single-cycle processor** using **Verilog**, encompassing a Control Unit, ALU, and memory unit.
 - Designed an Instruction Set Architecture, implemented the necessary control signals and integrated key components.
 - Conducted thorough testing by performing a bubble sort algorithm using instructions defined by its ISA.
- Project Mentor, AlgoArena, Programming Club IITK** (Apr'23-Jul'23)
 - Mentored a group of 20 juniors for the **Competitive Programming** based project organized by Programming Club.
 - Gave lectures on **Data structures and Algorithms** and various other techniques helpful in competitive programming.
 - Created topic-wise mashups and problem sets for complete practice and understanding of the topic.

Technical Skills

- Programming Languages:** C, C++, HTML, CSS, JavaScript, Python, Go
- Web Frameworks and Miscellaneous:** React.js, Next.js, Gin, $\LaTeX$, Git, Bash, Numpy, Pandas

Positions of Responsibility

- Secretary, Programming Club IITK** (Jan'23-May'23)
 - Among the 24 secretaries responsible for managing various activities, sessions, competitions pertaining to the club.
 - Authored and tested competitive programming problems on **Polygon** and **Codeforces** platforms for the **Freshers Programming Contest** organised for Y22. Additionally, wrote detailed editorials for the problems.
 - Conducted testing and provided editorial write-ups for the competitive programming based **Test Series** conducted by Programming Club, ensuring the quality and comprehensibility of the problems.

Relevant Courses

Data Structure and Algorithms	Fundamentals of Computing
Introduction to Real Analysis	Linear algebra and Differential Equations
Discrete Mathematics	Modern Cryptology
Software Development and Operations	Computer Organization
Introduction to Probability	Logic in Computer Science

Extra-Curricular Activities

- Received first postition in the **Traveller Transpiler** problem statement in **Takneek'23**, the annual intra-IITK Science and Technology Championship.